Inverted MAML Card 1: Round 16 Thermal Bit-Slicing Report

Hardware: Card 1 (T120+T20, 1k/2.2k) | **FW:** v2.2-MAML (0x20261030) | **Params:** 40% PWM, K=1, LR=0.055, L2=1920, Mom=0.85, Clamp=±100 LSB

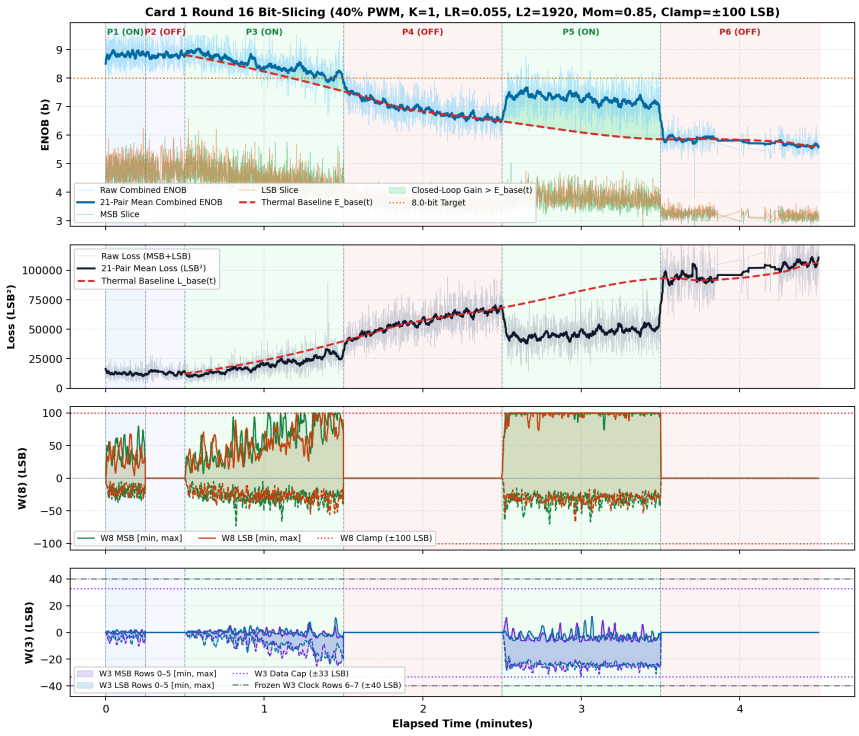
1. Round 16 Summary & Cumulative Comparison (R1–R15)

- **Cold Verification:** Phase 1 ON = **9.45 b**, Phase 2 OFF = **9.45 b** (W8=0 strictly).
- **Round 16 Detrended Trend Metrics:** Lift vs E_base = **+0.94 b**, Loss Ratio = **1.67x**, Climb = **+0.89 b**, Jitter = **0.40 b**, Recovery = **0.75**, Score = **+2.443**.

Rnd	LR	L2	Mom	Clamp	Lift	LossR	Climb	Jitt	Recov	Score
R8dc_fix	0.055	2048	0.85	±75	+1.16b	1.85x	+0.68b	0.42b	0.00	+1.855
R9dc_fix	0.055	1920	0.85	±75	+0.81b	1.59x	+0.66b	0.38b	0.50	+1.756
R10dc_fix	0.055	1920	0.85	±78	+0.96b	1.67x	+1.13b	0.40b	0.00	+1.898
R11dc_fix	0.055	1920	0.85	±80	+0.95b	1.70x	+0.98b	0.42b	0.00	+1.810
R12dc_fix	0.055	1920	0.85	±82	+1.04b	1.75x	+0.83b	0.42b	0.00	+1.721
R13dc_fix	0.055	1920	0.85	±85	+1.12b	1.78x	+1.06b	0.43b	0.50	+2.645
R14dc_fix	0.055	1920	0.85	±90	+0.78b	1.55x	+0.97b	0.39b	0.67	+2.117
R15dc_fix	0.055	1920	0.85	±100	+0.94b	1.67x	+0.89b	0.40b	0.75	+2.443

2. Phase-by-Phase Telemetry (3,821 Pairs)

Phase	Regime	N	Start->Late (Mean)	Δ vs E_base	Loss Start->Late	W8 Pk [M,L]	W3 Pk	W8 nz
P1	Cold / Regler ON	334	8.78->8.86 (8.83b)	-	13834->11622	[80,68]	10	334
P2	Cold / Regler OFF	265	8.89->8.79 (8.80b)	-	11010->12702	[0,0]	0	0
P3	Warm / Regler ON	1035	8.86->8.20 (8.48b)	+0.38 b	12074->24981	[100,100]	25	1035
P4	Warm / Regler OFF	867	7.62->6.62 (6.92b)	+0.02 b	38805->64460	[0,0]	0	0
P5	Warm / Regler ON	830	6.85->7.15 (7.26b)	+1.24 b	58534->49964	[100,100]	33	830
P6	Warm / Regler OFF	490	5.86->5.64 (5.78b)	-0.03 b	93826->105523	[0,0]	0	0



3. All W(8) (7x6 MSB+LSB) & W(3) (8x6 MSB+LSB) Weight Dynamics — Round 16

• **Full 180-Synapse Per-Pair Capture:** Every individual weight in **Matrix 8 MSB (42 weights)**, **Matrix 8 LSB (42 weights)**, **Matrix 3 MSB (36 active + 12 clock)**, and **Matrix 3 LSB (36 active + 12 clock)** is plotted below across all columns `Col 0..5`. W(8) enforces anti-windup clamping at ±100 LSB, while W(3) rows 0–5 stay within ±33 LSB so rows 6–7 clock staircases remain locked at ±40 LSB.

Round 16 Full Hardware Weight Telemetry: ALL 84 W(8) [7x6 MSB+LSB] & ALL 96 W(3) [8x6 MSB+LSB] Synapses

